

A primal heuristic to compute an upper bound set for multi-objective 0-1 linear optimisation problems

MODEM '21: 1st Multi-Objective Decision Making Workshop

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Xavier GANDIBLEUX (Université de Nantes, France)

Guillaume GASNIER (Université de Nantes, France)

Saïd HANAFAI (Université Polytechnique - INSA Hauts-de-France, France)

Context: notations, p -01LP

Multi-objective 0-1 linear optimisation problems (p -01LP):

$$\begin{aligned} \min \quad & z(x) = Cx \\ \text{s/t} \quad & Ax \leq b \\ & x \in \{0, 1\}^n \end{aligned}$$

where:

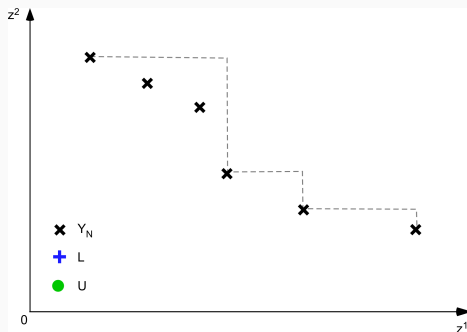
- $A \in \mathbb{R}^{m \times n}$,
the m constraints $A_i x \leq b_i$, $i = 1, \dots, m$ and $b \in \mathbb{R}^m$;
- $x \in \{0, 1\}^n$,
the vector of n binary variables x_j , $j = 1, \dots, n$;
- $C \in \mathbb{R}^{p \times n}$,
the objective matrix where $p \geq 2$;
- $X = \{x \in \{0, 1\}^n \mid Ax \leq b\} \subseteq \mathbb{R}^n$,
the set of feasible solutions, with $\mathbb{R}^n$ the decision space;
- $Y = \{Cx \mid x \in X\} \subseteq \mathbb{R}^p$,
the outcome set, with $\mathbb{R}^p$ the objective space.

Context: X_E and Y_N

Y_N is the set of nondominated points

X_E is a complete set of efficient solutions

- $y = z(x)$
- $y^* \in Y$ is **nondominated**, if $\nexists y \in Y \mid y_i \leq y_i^*, \forall i$ and $y \neq y^*$
- $x^* \in X$ is **efficient** if y^* is nondominated.



Context: p -01LP $\rightarrow$ 2-SPA

The bi-objective set partitioning problem (2-SPA) as benchmark:

$$\begin{array}{lll} \min & z(x) & = \quad Cx \\ \text{s/t} & Ax & = \quad 1 \\ & x & \in \quad \{0,1\}^n \end{array}$$

where

$$\begin{array}{l} C \in \mathbb{Z}^{p \times n} \\ A \in \{0,1\}^{m \times n} \end{array}$$

Y_N computed with the `vOptGeneric.jl` package of vOptSolver

Xavier Gandibleux, Gauthier Soleilhac, Anthony Przybylski, and Stefan Ruzika

vOptSolver: an open source software environment for multiobjective mathematical optimization.

IFORS2017: 21st Conf. of the Int. Federation of Operational Research Societies. July 17-21, 2017. Quebec City (Canada).

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Bound sets

Context of multi-objective branch and bound

Computing Y_N with a **MO branch and bound** algorithm:
an implicit enumeration principle viewed as a tree search.

Anthony Przybylski and Xavier Gandibleux.
Multi-objective branch and bound.
European Journal of Operational Research 260, 3 (2017), 856–872.

- L , a *lower bound set* on Y_N
- U , an *upper bound set* on Y_N

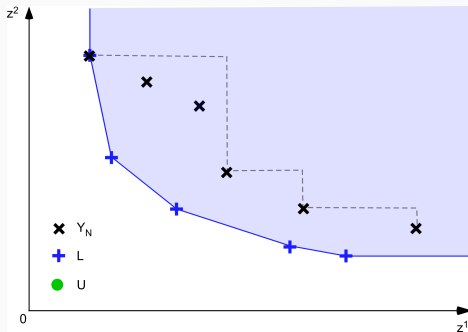
Matthias Ehrgott and Xavier Gandibleux.
Bounds and Bound Sets for Biobjective Combinatorial Optimization Problems.
Lecture Notes in Economics and Mathematical Systems, Vol. 507, 241–253, 2001.

Matthias Ehrgott and Xavier Gandibleux.
Bound sets for biobjective combinatorial optimization problems.
Computers & Operations Research 34 (2007), 2674–2694.

L and U bound sets

A lower bound set for Y' is a subset $L \subseteq \mathbb{R}^p$ such that

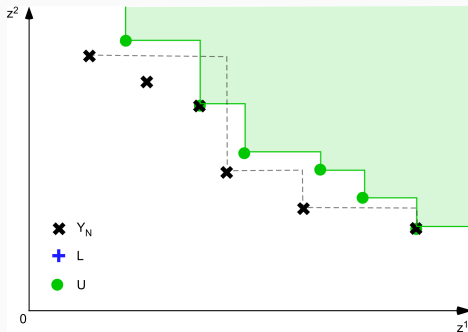
- For each $y \in Y'$ there is some $l \in L$ such that $l \leq y$,
- There is no pair $y \in Y'$, $l \in L$ such that y dominates l .



L and U bound sets

An upper bound set for Y' is a subset $U \subset \mathbb{R}^p$ such that

- For each $y \in Y'$ there is some $u \in U$ such that $y \leq u$,
- There is no pair $y \in Y'$, $u \in U$ such that u dominates y .



Summary

Our goal:

to tighten L and U over Y_N in order to prune as early as possible useless nodes in a branch-and-bound algorithm

The proposed algorithm:

devoted to the upper bound set U

The desirable characteristic:

U well representative of Y_N , and not necessary a close approximation of all $y \in Y_N$, which is a difference with the existing literature

Feasibility Pump

Feasibility Pump

A primal heuristic

- introduced in 2005 for computing feasible solutions of single-objective integer linear programs
- integrated in several MIP solvers such as GLPK, Gurobi, and CPLEX
- adapted in 2019 for MOP as competitor to NSGA-II (Deb et al. 2002), and Multidirectional Local Search (Tricoire 2012)

Matteo Fischetti, Fred Glover, and Andrea Lodi.

The feasibility pump.

Mathematical Programming 104(1):91–104, Sep 2005.

Timo Berthold, Andrea Lodi, and Domenico Salvagnin.

Ten years of feasibility pump, and counting.

EURO Journal on Computational Optimization, 7, 1–14 (2019).

Aritra Pal and Hadi Charkhgard.

A Feasibility Pump and Local Search Based Heuristic for Bi-Objective Pure Integer Linear Programming.

INFORMS Journal on Computing 31, 1 (2019), 115–133.

Feasibility Pump (Fischetti et al. 2005)

- **Initial solution:**

$\bar{x}^0 \in \bar{X}$, e.g.

$\bar{x}^0 := \arg \min \{c^T x : x \in \bar{X}\}$

$$\bar{X} = \{x \in [0, 1]^n : Ax \leq b\}$$

- Two sequences of solutions:

1) **binary:** $\tilde{x}^t \in \{0, 1\}^n$, $t \geq 0$

by a rounding procedure from $\bar{x}^t$

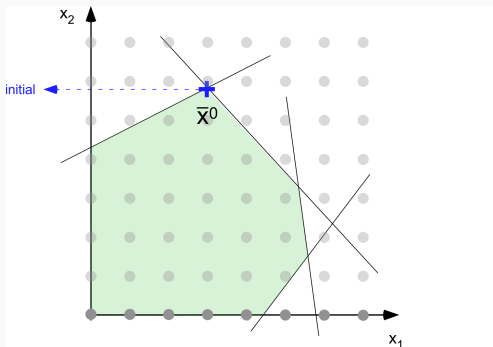
2) **fractional:** $\bar{x}^t \in \bar{X}$, $t > 0$

optimal solution of the projection problem

$\bar{x}^{t+1}$ is generated as the closest feasible LP solution with $\tilde{x}^t$

- Cycles: $(\tilde{x}^t, \tilde{x}^{t'})$, $t' > t$ repeated
→ PERTURBSOLUTION on $\tilde{x}^{t'}$

- Stopping conditions: **feasible solution** or maxTime limit reached



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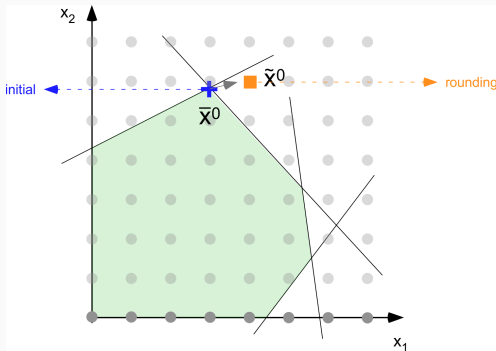
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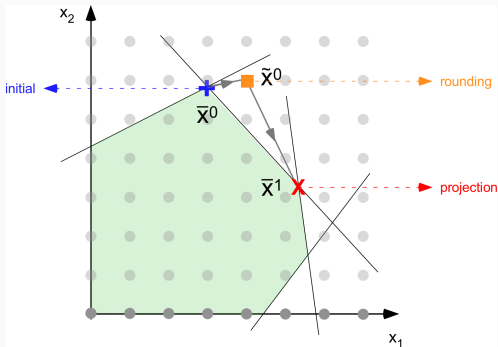
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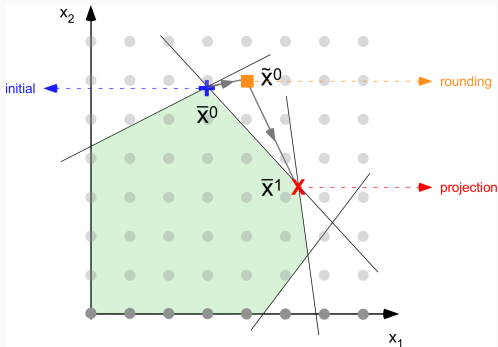
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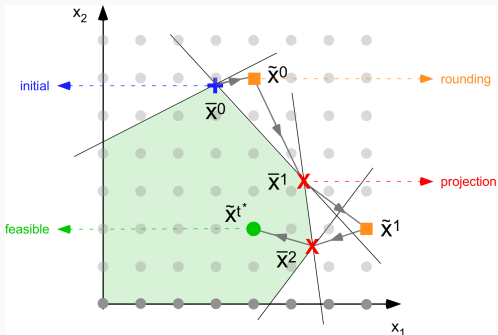
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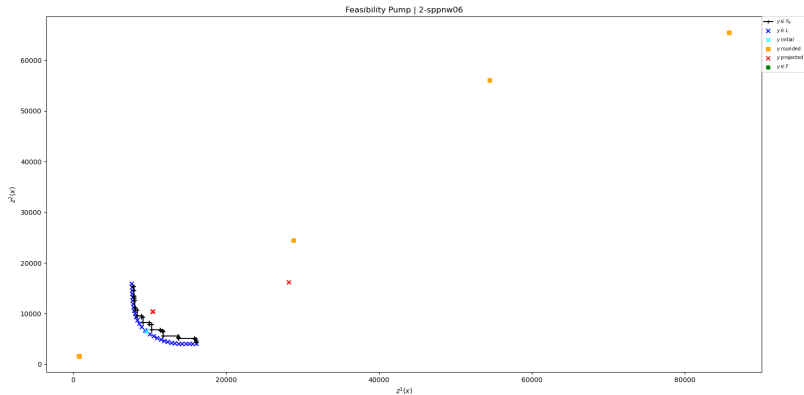
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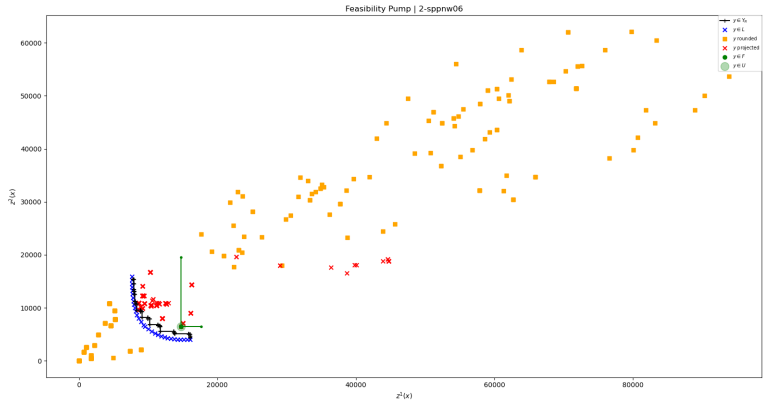
Feasibility Pump (2005): observation

E.g. for instance sppnw06:



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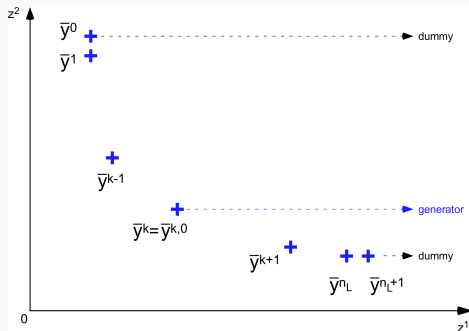
E.g. for instance sppnw06:



Gravity Machine

Gravity Machine

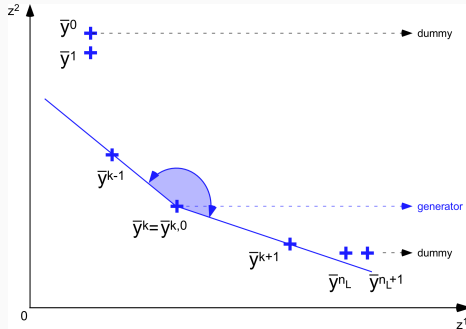
1/4: A set of generators $(\bar{x}^k, \bar{y}^k)$ with $\bar{y}^k := z(\bar{x}^k)$:



ϵ -constraint LP resolutions

Gravity Machine

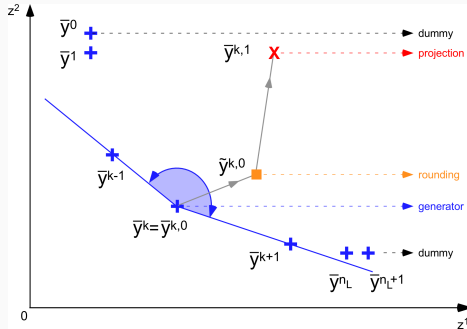
2/4: A cone open to the search area in objective space:



Hard-constraints over $\bar{y}^{k,t}, t = 0$

Gravity Machine

3/4: Alternance of rounding/projecting operations:

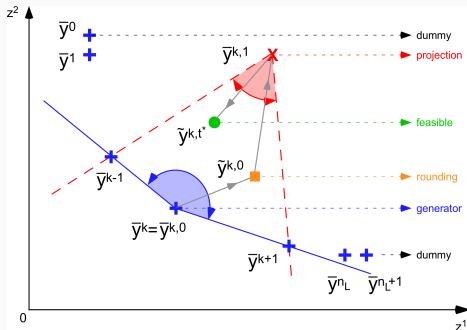


$$\tilde{x}^{k,t}, H, \text{cycle} \leftarrow \text{ROUNDINGSOLUTION}(\bar{x}^{k,t}, k, H)$$

and

$$\bar{x}^{k,t+1} := \arg \min \{ \Delta(x, \tilde{x}^{k,t}), x \in \bar{X} \}, t \geq 0$$

4/4: A cone open to the lower bound set in objective space:



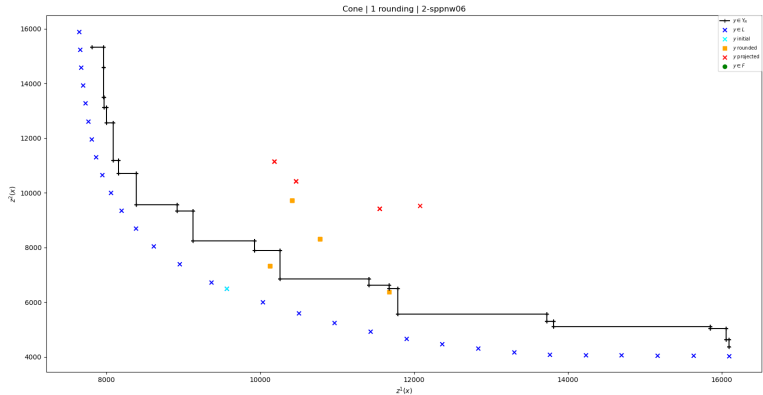
Soft-constraints over $\bar{y}^k,t, t > 0$

Gravity Machine: outline of the algorithm

```
Data:  $\mathcal{D}$ ,  $n_L$ , maxTrial, maxTime  
Result:  $L$ ,  $U$   
 $L, F \leftarrow \text{COMPUTEGENERATORS}(\mathcal{D}, n_L);$   
 $H \leftarrow \emptyset;$   
forall  $z(\bar{x}^k) \in L, \bar{x}^k \notin F$  do // each  $k$ -th unfeasible gen.  
    timeStart  $\leftarrow$  TIME(); trial  $\leftarrow$  0;  
    timeout  $\leftarrow$  false; feasible  $\leftarrow$  false;  
     $\tilde{x}, H, \text{cycle} \leftarrow \text{ROUNDINGSOLUTION}(\bar{x}^k, k, H);$   
    while  $\neg \text{feasible} \ \&\& \ \neg \text{timeout}$  do  
        trial  $\leftarrow$  trial + 1 ;  
         $\tilde{x}, F, \text{feasible} \leftarrow \text{PROJECTSOLUTION}(\tilde{x});$   
        if  $\neg \text{feasible}$  then  
             $\tilde{x}, H, \text{cycle} \leftarrow \text{ROUNDINGSOLUTION}(\tilde{x}, k, H);$   
            if cycle then  
                 $\tilde{x} \leftarrow \text{PERTURBSOLUTION}(\tilde{x});$   
            end  
        end  
        timeout  $\leftarrow$  (time()-timeStart  $\geq$  maxTime) || (trial =  
            maxTrial)  
    end  
end  
 $U \leftarrow \text{EXTRACTNONDOMINATEDPOINTS}(F);$   
Algorithm 1: Outline of « Gravity Machine »
```

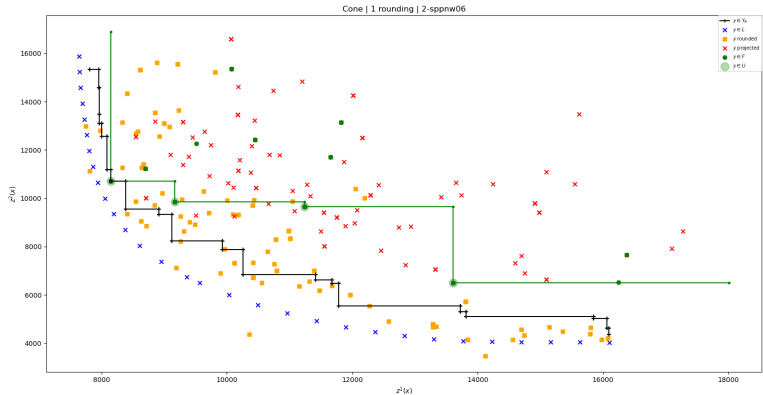
Gravity Machine: observation

E.g. for instance sppnw06:



Gravity Machine: observation

E.g. for instance sppnw06:



Numerical experiments

Environment

Program:

- Algorithm coded in Julia language (1.6)
- Use the algebraic modeling language JuMP (0.21.8) + GLPK (4.65)

Numerical instances:

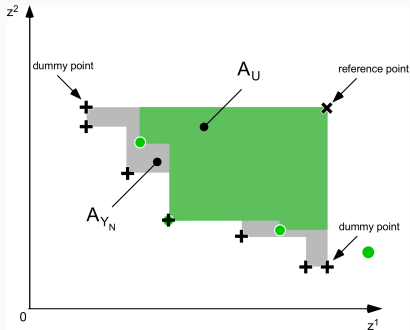
- 44 instances from the OR-library
<http://people.brunel.ac.uk/~mastjjb/jeb/orlib/sppinfo.html>
- Second objectives have been generated as follows:
 $c^2 := \text{randomShuffle}(c^1)$
- Available on the vOptLib (1998-):
<https://github.com/vOptSolver/vOptLib/tree/master/SPA>

Computer:

- Intel(R) Core(TM) i7-10870H processor running at 2,20 GHz
- 16 Go of RAM DDR4

Quality measure

The exclusive hypervolume contribution of solutions:



$$r = \frac{A_U}{A_{Y_N}}$$

Observing a single run

Parameters:

- $n_L = 30$
- `maxTrial` = 5
- `maxTime` = ∞ .

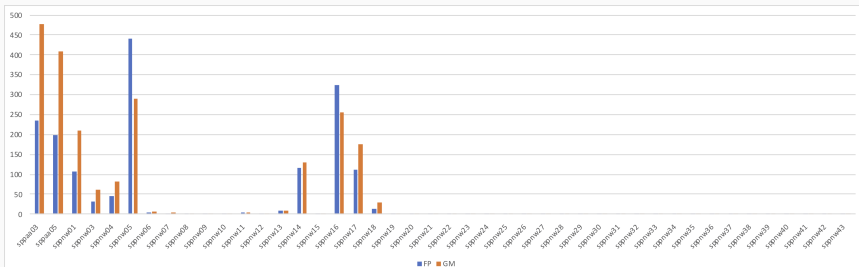
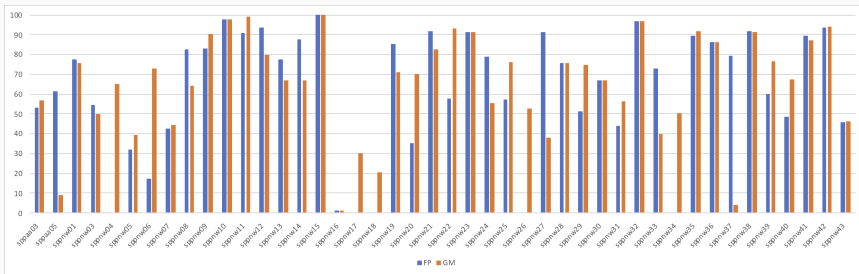
Measures:

- r the quality measure of U
- `CPUt` (sec),

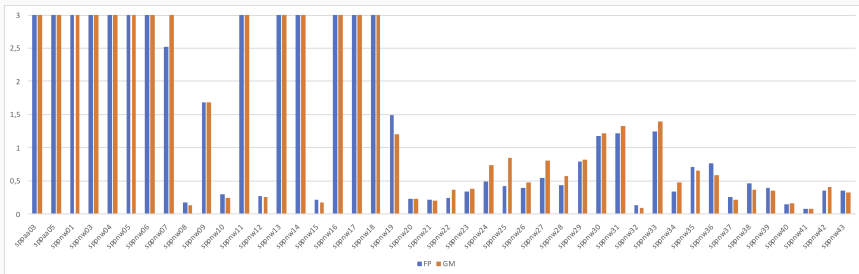
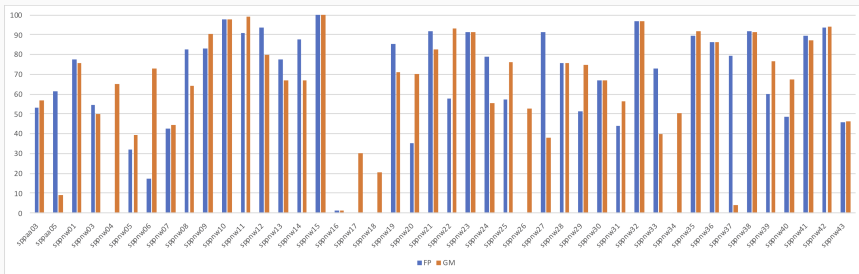
Discussion:

- Feasibility pump* (FP)
- Gravity Machine (GM)

Measures: top / ratio (%); bottom / CPUt (s)



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Ratio

<i>best</i>	/44
<i>FP</i>	16 (36.4%)
GM	21 (47.6%)
=	07 (16.0%)

CPUt

<i>best</i>	/44
FP	27 (61.5%)
<i>GM</i>	15 (34.0%)
=	02 (04.5%)

Ratio & CPUt

<i>best</i>	/44
<i>FP</i>	08 (18.2%)
<i>GM</i>	04 (09.1%)
none	32 (72.7%)

Conclusion

- easy to explain/understand, few parameters
- first results...and they are encouraging
- many potential search directions from now; e.g.:
 - using the objectives during the projection (c.f. Objective FP)
 - performing a local search on points belonging to U
 - etc.
- intensive numerical experiments:
 - impact of stochastic components
 - comparison with metaheuristics
 - etc.

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The source code (soon):

<https://github.com/xgandibleux/GravityMachine>

The datasets:

<https://github.com/vOptSolver/vOptLib/tree/master/SPA>